

Targeted-CNV-Learner: A Machine Learning Framework for Enhanced CNV Detection in Targeted Sequencing

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Motivation

Problem Statement: Detecting Copy Number Variations (CNVs) in targeted sequencing data is crucial for diagnosing a variety of genetic diseases. However, due to the limited genomic coverage in targeted panels, existing methods struggle to detect large or complex rearrangements, often resulting in high false-positive rates, which in clinical settings requires orthogonal validation increasing both personnel and resource costs.

Solution: Targeted-CNV-Learner is a novel machine learning (ML) framework designed to improve CNV detection accuracy by integrating the outputs of multiple CNV detection algorithms. By using a ML approach, the framework enhances precision and recall, particularly in clinical settings where targeted sequencing panels are commonly used.

Objectives

This project aims to develop and validate the Targeted-CNV-Learner framework, with the following key objectives:

- Maximize CNV detection recall:** Ensure accurate detection of most CNVs for reliable clinical diagnostics.
- Minimize false positives:** Reduce the need for costly follow-up validations.

Achieving a balance between precision and recall is essential to creating a dependable diagnostic tool that enhances clinical workflows and reduces operational costs.

Methodology

The methodology involves a systematic pipeline, moving from data acquisition to model development and validation:

- Data Acquisition:** A minimum of 200 samples, all processed using the same targeted sequencing panel, were collected. The dataset comprises BAM files (aligned sequencing reads) and corresponding BED files (defining the genomic regions of interest).
- Quality Assessment of Samples:** Principal Component Analysis (PCA) is performed on read-depth profiles to filter out low-quality samples.

- Insertion of In Silico CNVs:** Synthetic CNVs are inserted into the samples to enrich the training dataset and create a balanced example set for model learning.
- Running CNV Detection Algorithms:** Three CNV detection algorithms—GATK gCNV, DECoN, and GRAPES—are run on all samples. An overlap analysis identifies consensus CNVs, and detected CNVs are cross-referenced with the synthetic CNVs to classify them as true or false positives.
- Data Extension:** Additional features are extracted—including genomic context, sample quality metrics, and CNV detection quality features—to build a rich dataset for the machine learning model.
- ML model selection:** Random Forest and XGBoost models are trained on the extended dataset. The user can choose the best model based on performance metrics (e.g., accuracy, precision, recall).
- Feature selection and decision threshold adjustment:** To enhance model efficiency and interpretability, a feature selection process is employed, reducing dimensionality while retaining essential predictive power. Subsequently, the decision threshold is optimized to achieve an optimal trade-off between precision and recall, tailored to the specific needs of the user.

Finally, the framework also includes a module for **validating the model using samples containing real CNVs** (blue boxes in Figure 1).

Results

The Targeted-CNV-Learner framework was evaluated on a dataset of 400 samples using the SUDD147 genetic panel, designed for cardiovascular disease research. The model was validated using both in silico CNVs and real CNV datasets, with performance metrics compared against standalone algorithms.

Results on In Silico CNVs (Test Set):

Algorithm	Recall	Specificity	Precision	Accuracy
Decon	0.976	0.279	0.419	0.526
Grapes	0.851	0.726	0.623	0.766
GATK gCNV	0.841	0.859	0.761	0.853
Targeted-CNV-Learner	0.952	0.959	0.925	0.955

Table 1. CNV detection algorithm metrics over test set (20% in silico dataset)

Targeted-CNV-Learner achieved the best balance between high recall (95.2%) and specificity (95.9%) compared to other methods, reducing false positives significantly.

Results over Real CNVs:

The validation dataset comprised 41 experimentally validated CNVs:

Algorithm	TP	FP	TN	FN
Decon	38	213	39	3
Grapes	40	31	221	1
GATK gCNV	40	47	205	1
Targeted-CNV-Learner	40	31	221	1

Table 2. CNV detection algorithm metrics over dataset containing real CNVs.

Targeted-CNV-Learner correctly identified 40 CNVs (97.56% recall), matching GRAPES and GATK gCNV. However, both Targeted-CNV-Learner and GRAPES had fewer false positives (31), outperforming GATK gCNV and demonstrating superior specificity and reliability.

Drawbacks and limitations

While the performance of Targeted-CNV-Learner is promising, there are several important limitations to consider:

- Small real CNV validation dataset.
- Tested only on SUDD147 genetic panel.
- Requires a minimum of 200 samples.
- High computational burden due to multiple CNV detection algorithms.

Conclusion

The Targeted-CNV-Learner framework addresses key limitations of existing CNV detection methods, significantly improving precision and specificity over standalone algorithms. Its adaptability to various gene panels, combined with a lower false positive rate and high recall, establishes it as a valuable advancement in clinical genomics, enabling more efficient and accurate CNV diagnosis in research and clinical settings.



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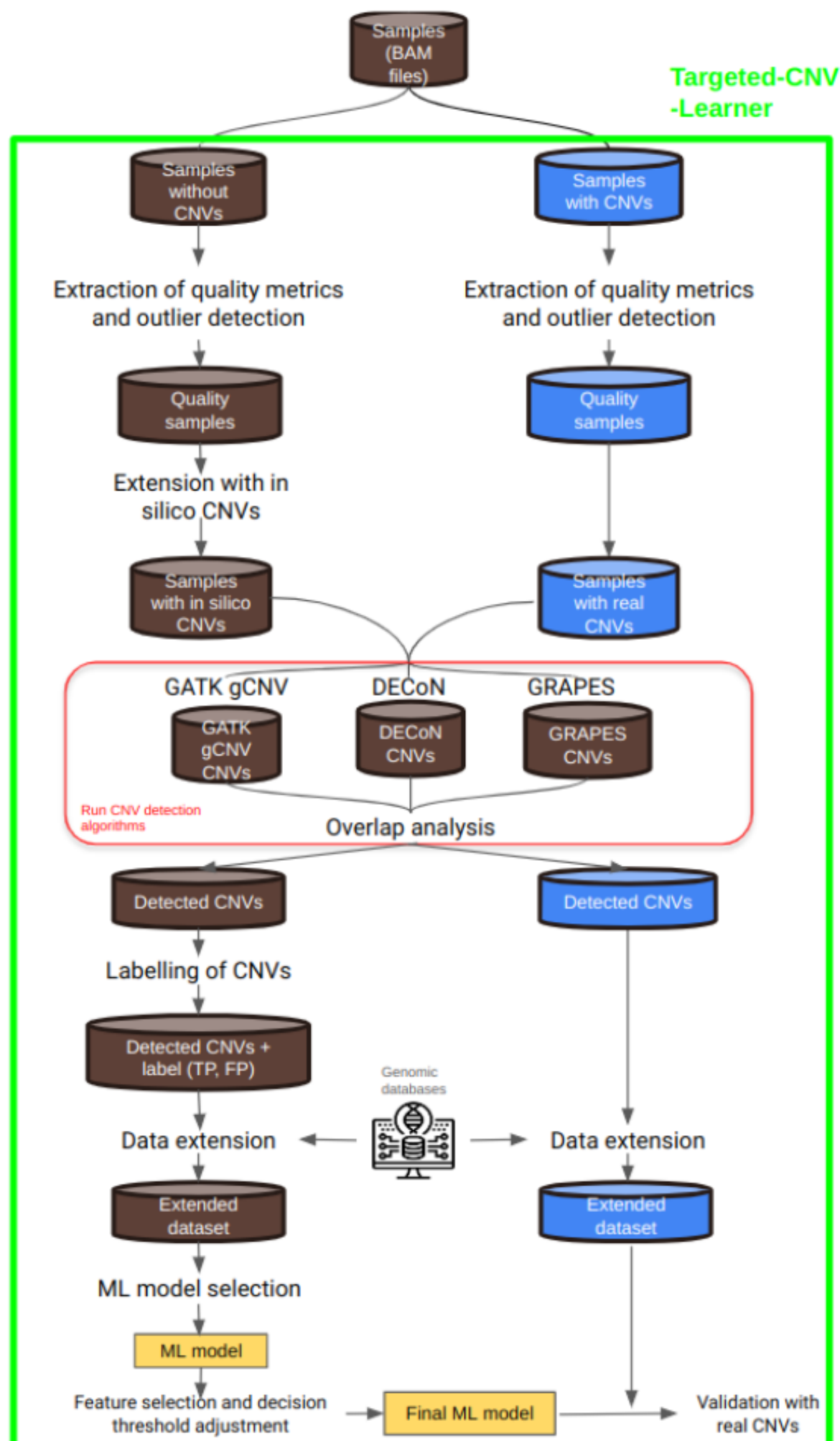


Figure 1. Overview of the methodology applied in this study.